Great Experiences – President Worthen

Internship for Credit Course

Internships for Credit at BYU

- Make students better qualified for being hired [Industry Advisory Board]
- Require students and employers to reflect on their experience and its importance
- Require employers to agree to education-focused internships
- Have yielded overwhelmingly positive reports from students who participated in the pilots
- Shift mentoring workload to employer [one adjunct per 100 students per semester]
- Helps students overcome imposter syndrome

Internship for Credit Learning Outcomes

- Gain experience in
  - a software engineering team
  - using software engineering design and implementation
  - software development in meaningful work environment
- Refine understanding of career goals and preferences
- Increase professional network

Internship for Credit Learning Activities

- Perform duties required by sponsoring company
- Complete a two-page mid-internship and two-page final reflective learning report
- Complete a survey of the internship experience.

Question

Why give credit for something students already do?

Can we provide a better experience with our internship for credit course than what students get if they did it on their own? Is it better enough to replace a technical elective?

How does this compare to a capstone, or mentored research, experience?

Response

CS Internship for Credit

- Deep Experiential Learning (DEL) has more and longer-lasting positive effects on students than traditional classes
- DEL is part of the BYU mission [President Worthen]
- It is desirable to give every CS student at least one DEL
- DEL is labor intensive

Internships and Scaffolded Reflection* (SR)

- Students who have an Internship with SR
  - Are more positively disposed to industry and school
  - Increase awareness of the importance of interpersonal relationships and life-long earning
  - Increase maturity and sense of direction
  - Become more thoughtful and deliberate in their studies
  - See connections between their courses and Computer Science practice better
  - Increase enthusiasm for their education

- Minnes, Mia, Sheena Ghanbari Serslev, and Madison Edwards. "Attitude shifts and transformation during Computer Science and Engineering Student Internships." *2020 ASEE Virtual Annual Conference Content Access*. 2020.

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Internships for credit are a “force multiplier” that make their other CS classes more impactful and increase their appreciation for their BYU education

- Minnes, Mia, Sheena Ghanbari Serslev, and Madison Edwards. "Attitude shifts and transformation during Computer Science and Engineering Student Internships." *2020 ASEE Virtual Annual Conference Content Access*. 2020.

Background for the motivated

History of Three Pilot Courses

- Course taught by adjunct Bradley Baird
  - Spiel about Bradley?
- Spring/summer 2021: 15 students (capped at 15)
- Spring/summer 2022: 42 students (capped at 50)
- Spring/summer 2023: 22 students so far (capped at 50)

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Companies Participating (all signed BYU agreements)

This is what I have so far for this year and for last (Duplicates removed)

2023	2022
Arctech Charge	Air Force Civilian Service
BambooHR	Amazon
Brevium	Apple
BYU	AT&T
BYU OIT	Awardco
CHG Healthcare	BYU Broadcasting
Everee	BYU Center for Teaching
Expedia Group	Canyonwall LLC
LDS Church	Clearwater Analytics
Jewelry TV	Clever Coding LLC
FamilySearch	Enium Capital Group
Northrop Grumman	Famiy Search
Nu Skin	Fidelity Investments
PAXI Smart	goldman sachs
Qualtrics, LLC	Pattern
T. Rowe Price	Pluralsite

Deep Learning is Different from Classroom Instruction

George D. Kuh – coprincipal investigator of the National Institute of Learning Outcomes Assessment (reported by President Worthen)

- Deep learning is different from normal learning
  - Not only acquire information but …
  - Understand the full meaning of and …
  - The theory underneath the information
- High impact practices
  - All but one high impact practice took place outside of the classroom
  - Experiential learning

Great Experiences – Finding a Meaningful Career

"… asking students who have graduated, who have found themselves in a job or career that they found particularly meaningful … What was going on in your undergraduate career that kind of indicated that you might be in this meaningful, in this place of finding great meaning in your vocation and in your employment … The Big Six" (https://experience.byu.edu/experiential-learning-cycle-definition-examples-and-assessment, accessed 24 Jan 2023)

1. A professor who excited me about learning.
2. Professors who cared about me as a person.
3. A mentor who helped me pursue my goals and dreams
4. Work on a project that took a semester or more to complete.
5. Internship or job that allowed me to apply my learning.
6. Extremely active in extracurricular activities and organizations.

Reflective Learning – Educational Value in an Internship

Career Preparation

"Internships and apprenticeships stand out as the applied learning experience most highly valued by employers, 93 percent of executives, 94 percent of hiring managers say they will more likely hire a recent grad who held an internship/apprenticeship within a company or organization…. While both audiences value applied experiences and real world skills, only 33% of executives and 35% of hiring managers think that recent graduates are well-prepared to apply knowledge and skills and real-world settings."(American Association of Colleges and Universities' employer survey from 2018). From https://experience.byu.edu/experiential-learning-cycle-definition-examples-and-assessment, accessed 24 January 2023

Quality Control – Shaping the Internship Experience

Community Building – Belonging and Alumni Relations

A study performed by the BYU Alumni Association explored what caused people to join alumni chapters and donate to BYU. Feelings of belonging in a Department were strongly associated with alumni activity. Feelings of belonging were associated with

- Being part of a BYU-sponsored team, cast, or group
- Being mentored by an individual faculty member
- Receiving career and internship placement support from a department

CS 199R – Internship

Prerequisites and Hour Requirements

I. **Prerequisites**
   A. Eligible applicants are juniors in the program, starting the senior year, and expected to be eligible for full-time employment in the following year
   B. Student has received approval for internship by CS Internship Coordinator
   C. Students bring their own internship to the course

II. **Hours**
   A. Credit: 3 hours
   B. Work Experience: 150 hours
   C. Pre-internship meeting: 1 hour
   D. Reporting: 3 hours

Learning Outcomes

A. Student gains experience in a software engineering team
B. Student gains experience in using software engineering design and implementation
C. Student gains experience in software development in meaningful work environment
D. Student refines understanding of career goals and preferences
E. Student increases professional network

Learning Activities – During Internship

A. Perform duties required by sponsoring company
B. Complete a two-page mid-internship report (12 pt., single spaced) containing four or more of the following
   1. Description of company culture and practices
   2. Skills required for success in the company
   3. Alignment of preparation from courses and job responsibilities
   4. Frustrating or disappointing aspects of the internship
   5. Surprising or rewarding aspects of the internship
   6. Specific skills developed
   7. Needs for specific skill development

Learning Activities – Reflective Learning

A. Complete a two-page final report (12 pt., single spaced) with two parts:
   1. A brief description of what you did for your internship
   2. A report on what you learned about:
      a) Software engineering patterns and practices
      b) Principles and practices from CS courses used (or not used) in your internship
      c) Being part of a software development team
      d) Company culture and mentoring
      e) Personal career goals and desires

B. Complete a survey of the internship experience. Questions include
   1. Hours spent per week
   2. Quality of mentoring
   3. Quality of software produced
   4. Company culture including diversity and respect
   5. Alignment of internship experience with educational/career goals

Learning Environment

I. **Sponsor Activities**

A. Complete BYU Internship agreement (key phrases)

1. An internship is a cooperative student program between BYU and the Experience Provider. The Experience Provider will provide supervision, facilities, and instruction that help students of BYU (each an “Intern”) acquire skills and knowledge related to their chosen field of study or occupation.
2. Each Intern is placed with the [Intership] Provider in order to receive educational experience as part of the academic curriculum; duties performed by an Intern are not performed as an employee of the [Internship] Provider but rather in fulfillment of the academic requirements of the educational experience and are performed under direct supervision by the Experience Provider’s Rev. 1 November 2022 personnel.

B. Internship sponsors will complete a midterm report and submit to instructor

1. Team composition (size, role performed, etc.)
2. Nature of work assigned
3. Team contribution
4. Strengths shown by Intern
5. Areas for growth

C. Internship sponsors will complete an end of semester survey.

Report from the Two Year Experiment: Internship for Credit

See attached file

Internships for Credit at BYU

Sample of “Similar” Departments

Department	Internship for Credit?	Capstones?
Statistics	Yes	Data science emphasis
ACME		
Mechanical Engineering	No	All students
Electrical and Computer Engineering		All students
Chemistry		

Internships for Credit at Other Institutions

Sample of Other Universities

Department	Internship for Credit?	Capstones?
University of Washington	Yes	
Stanford	No	
U. Georgia School of Computing	Yes	